

Microsoft Project 2016 Step by Step Manual

Introduction

Microsoft Project 2016 (MS Project) Step by Step is designed for use as a learning and reference resource by students who want to use MS Project to create and manage engineering projects more efficiently. This Step by Step manual has been designed so you can read it to learn about MS Project 2016 and then build your skills as you learn to perform increasingly specialized procedures.

It provides a brief overview of the user interface and covers creating and linking tasks, adding resources and assigning them to tasks, and scheduling.

Overview of the interface

Like the other Microsoft Office 2016 applications, MS Project 2016 incorporates the **Ribbon**, **Quick Access toolbar**, **backstage view**, and **context-sensitive menus and toolbars**. These all make working with Project 2016 more familiar and intuitive.

When you start MS Project 2016, its **Start** screen appears (See figure 1). Here you can quickly open a plan that was recently opened, open another plan, or create a new plan. The new plan can be empty or it can be based on a template or on another plan.

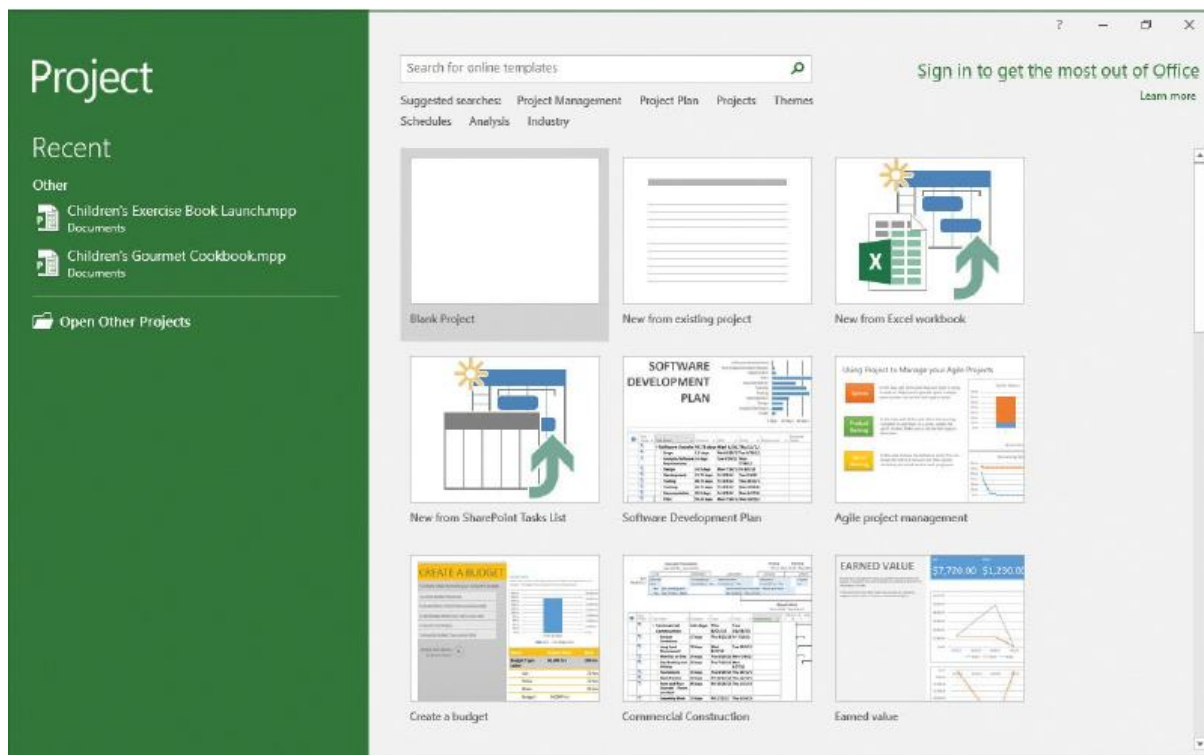
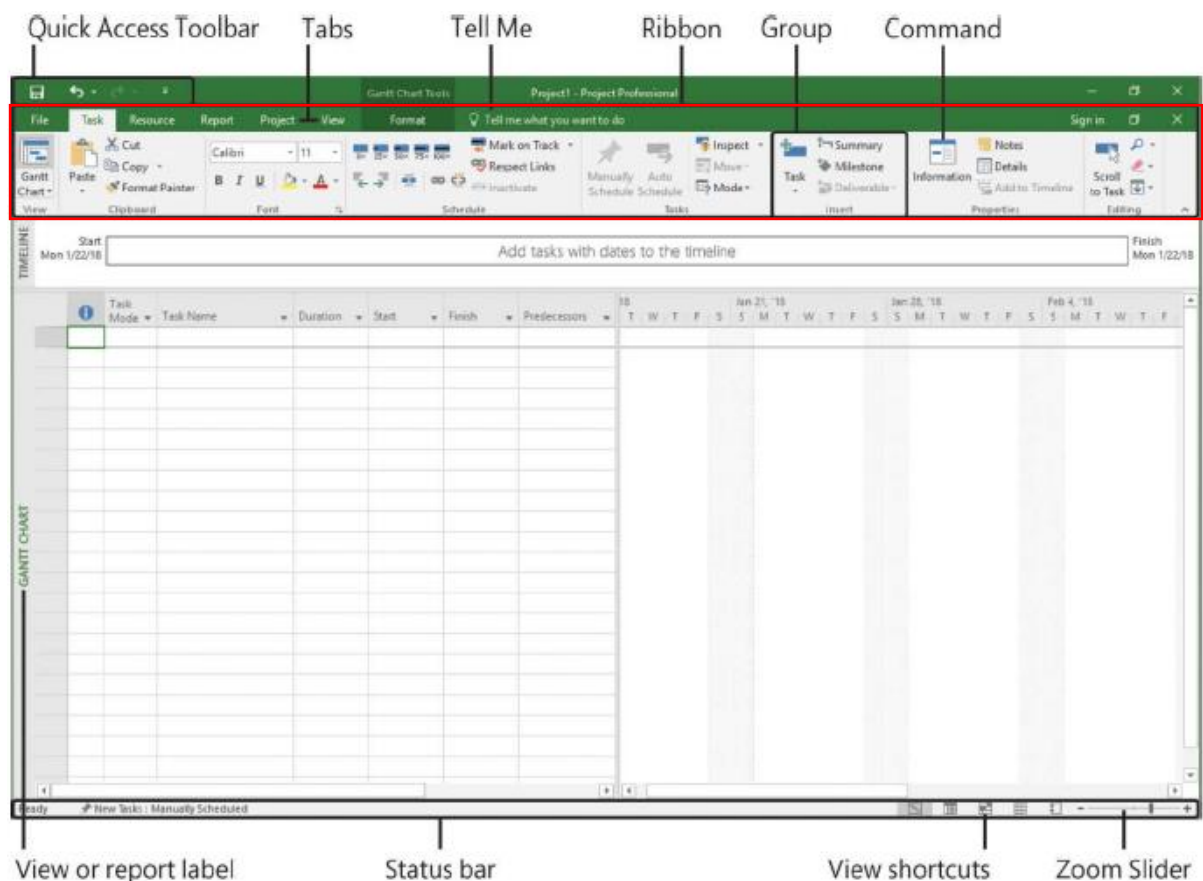


Figure 1: Project 2016 Start Screen

The default view in Project 2016 is the **Gantt Chart** view (see Figure 2). It has two main areas: the **Task Table** (left side of the view) and the **Timescale** (right side of the view). The **Task Table** lists the project tasks and their accompanying information. The **Timescale** displays task bars that correspond to the tasks listed in the task table. The **Task Bars** show the duration of each task, from their planned start date to their planned finish date, over time. You can move the separator bar between the timescale and the task table to the left or right to display more of either area. The timeline provides a graphical view of a project along a single left-right axis.



Here are the major parts of the Project interface; note the label of the active view along the left edge

Figure 2: MS Project 2016 Project interface

Creating a new plan as a blank project

1. Click the **File** tab to display the backstage view.
2. Click **New** to display the new page.
3. Click **Blank Project**.

To set or change the plan's start date

1. On the Project tab, in the Properties group, click **Project Information**.
2. In the Project Information dialog box, in the **Start Date box**, enter the start date you want or click the **Arrow** to select one from the calendar.
3. Click **OK** to accept the start date and close the Project Information dialog box.

To save the new plan

1. Click the **File tab**, and then click Save As.
2. On the **Save As page**, navigate to the location where you want to save the plan.

Set nonworking days in the project calendar

Calendars are the primary means by which you control when each task and resource can be scheduled for work in Project. In this section we focus on the project calendar.

The project calendar defines the general working and nonworking days and times for tasks. Project includes multiple calendars, called base calendars, any one of which can serve as the project calendar for a plan. You select the base calendar that will be used as the project calendar in the Project Information dialog box as shown in Figure 3.

Project Information for 'Project1'

Start date: Mon 1/8/18 Current date: Mon 1/8/18

Finish date: Mon 1/8/18 Status date: Mon 1/8/18

Schedule from: Project Start Date Calendar: Standard

All tasks begin as soon as possible. Priority:

Enterprise Custom Fields

Department:

Custom Field Name	Value
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Set the plan's start date, project calendar, and other essential schedule settings in the Project Information dialog box

Figure 3: Setting non-working days in the calendar

The Calendar list contains the three base calendars that are included with Project:

Only one of the base calendars serves as the project calendar; the Standard calendar is the default.

Think of the project calendar as your organization's normal working days and hours. For example, this might be **Monday** through **Friday, 8:00 A.M.** through **5:00 P.M.**, with a one-hour lunch break each day. Your organization or specific resources might have exceptions to this normal working time, such as holidays or vacation days.

You customize calendars in the **Change Working Time** dialog box (which opens when you click the Change Working Time button on the Project tab).

The selected day's working times and calendar

Change Working Time

For calendar: **Standard (Project Calendar)**

Calendar 'Standard' is a base calendar.

Legend:

- ☐ Working
- ☐ Nonworking
- 31** Edited working hours
- 31** Exception day
- 31** Nondefault work week

Click on a day to see its working times:

January 2018

S	M	T	W	Th	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

Working times for January 8, 2018:

- 8:00 AM to 12:00 PM
- 1:00 PM to 5:00 PM

Based on:
Default work week on calendar 'Standard'.

Exceptions | **Work Weeks**

Name	Start	Finish

Buttons: Help, Options..., OK, Cancel, Details..., Delete

Customize a calendar's working time in the Change Working Time dialog box

Figure 4: Customising calendar to suit your working days and time

To select the project calendar

1. On the **Project tab**, in the **Properties group**, click the **Project Information button**.
2. In the Project Information dialog box, in the **Calendar box**, click the **Arrow**, and then click the calendar you want to use as the project calendar.

To set a specific date as nonworking

1. On the **Project tab**, in the **Properties group**, click the **Change Working Time button**.
2. On the **Exceptions tab** in the lower portion of the **Change Working Time dialog box**, in the **Name field**, enter a description of the **Exception**. In addition, note that you do not need to name calendar exceptions, but it is a good practice for you or others to identify the reason for the exception.
3. In the **Start** and **Finish fields**, enter or select the dates you want.
4. Click **OK**.

Working with new tasks

Most projects begin with a list of the tasks that need to be completed. Once you create your task list, you can then define the relationships between the tasks. Tasks can be automatically or manually scheduled, depending on the needs of the project and the information currently available. At any point during a project, you can change a task from manual scheduling to automatic scheduling, or vice versa.

Creating a Manually Scheduled Task

By default, all new tasks in Project 2016 are set for manual scheduling. With manual scheduling, you can fully control the task's duration, start date, and finish date. The duration can be entered in minutes (m), hours (h), days (d), weeks (w), or months (mo). Manually scheduled tasks are marked with the pushpin icon in the Task Mode field.

To create a manually scheduled task:

1. In the first row of the task table, enter Website Ideas and Mock-ups in the Task Name Field (see Figure 5).
2. Enter 1w in the Duration field. This sets the duration of the task to one week.
3. Enter Monday in the Start field. Project 2016 automatically calculates the finish date using the given information and fills in the Finish field. In addition, a task bar appears in the timescale (on the same row as the task) showing the task's time frame.

	Website Ideas and Mockups	1 wk	Mon 8/25/14	Fri 8/29/14
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Figure 5: Manually scheduled Task in a Task Table

Creating an Automatically Scheduled Task

With automatically scheduled tasks, you can have Project 2016 calculate dates and durations. Automatically scheduled tasks are marked with the Gantt bar icon in the Task Mode field.

To create an automatically scheduled task:

1. In the second row of the task table, enter Preparation and Planning in the Task Name Field (see Figure 6).
2. Click in the Task Mode field, click the down arrow that appears, and then click Auto Scheduled.

By default, Project 2016 assigns the task an estimated duration of one day and automatically fills in the Duration, Start, and Finish fields. In addition, a task bar appears in the timescale showing the task's time frame.

3. To change the duration of the task to two days, enter 2d in the Duration field. Project 2016 recalculates the finish date and updates the Finish field.

	Preparation & Planning	2 days	Thu 8/21/14	Fri 8/22/14
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Figure 6: Automatically scheduled Task in a Task Table

Linking Tasks

When you link tasks, you create a relationship between them called a task dependency. Project 2016 supports four types of task dependencies. By default, Project 2016 creates finish-to-start links, which means the first task needs to finish before the second task can start. See figure 7 to other task links:

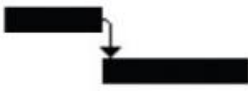
Task relationship	Meaning	Appearance in the Gantt chart	Example
Finish-to-start (FS)	The finish date of the predecessor task determines the start date of the successor task. This is the default task relationship.		A book chapter must be written before it can be edited.
Start-to-start (SS)	The start date of the predecessor task determines the start date of the successor task.		The tasks <i>Schedule Prepress Time</i> and <i>Order Paper</i> are closely related, and they should start simultaneously.
Finish-to-finish (FF)	The finish date of the predecessor task determines the finish date of the successor task.		Tasks that require specific equipment must end when the equipment rental period ends.
Start-to-finish (SF)	The start date of the predecessor task determines the finish date of the successor task.		The time when the print run is scheduled to start determines when a binder selection task must end.

Figure 7: Different Tasks Links

To Link Task

1. Select the **Names** of the tasks you want to link. If the tasks are not adjacent, select the first task, hold down the Ctrl key, and then select the additional task or tasks.
2. On the **Task tab**, in the Schedule group, click the **Link the Selected Tasks button** (which looks like two chain links). In addition, note that Project 2016 links the tasks with a finish-to-start relationship. See figure 8 below:

Change highlighting indicates values that are affected after a change is made to a plan

	Task Mode	Task Name	Duration	Start	Finish	Predecessors	
1		Planning Phase	2 days	Mon 1/8/18	Tue 1/9/18		
2		Assign launch team members	1 day	Mon 1/8/18	Mon 1/8/18		
3		Design and order marketing	Check with Marketing	Tue 1/9/18	Tue 1/9/18	2	
4		Planning complete	0 days				
5		Public Launch Phase	6 days	Mon 1/22/18	Tue 1/30/18		

A link line indicates a task relationship between tasks

Create schedule dependencies between tasks by linking them

Figure 8: Linking tasks

To unlink tasks

1. Select the **Tasks** you want to unlink.
2. On the **Task tab**, in the **Schedule group**, click the **Unlink Tasks button** (which looks like a broken chain link).

To force a manually scheduled task to respect its predecessor task's scheduling result

1. Select the manually scheduled task you want to be rescheduled as determined by its predecessor task relationships.
2. On the Task tab, in the Schedule group, click the Respect Links button

Create summary tasks or task Grouping

To promote a task to a summary task

1. Select the tasks directly below the task that you want to make a summary task.
2. On the **Task tab**, in the **Schedule group**, click the **Indent Task button**.

	Task Mode	Task Name	Duration	Start	Finish	Predecessors	
1		Assign launch team members	1 day				
2		Design and order marketing material	Check with Marketing				
3		Planning complete	0 days				
4		Public Launch Phase	6 days	Mon 1/22/18	Tue 1/30/18		
5		Distribute advance copies	2 days				
6		Coordinate magazine feature articles	6 days	Mon 1/22/18	Tue 1/30/18		
7		Launch public web portal for	About two weeks				

Subtasks indented under the summary task

Project uses text and bar formatting to distinguish between summary tasks and subtasks

Figure 9: Creating a task summary

To insert a new summary task within a task list

1. Select the name of the tasks that will become subtasks.
2. On the **Task tab**, in the **Insert group**, click **Summary**.

Project inserts a row for a new task, indents all of the selected tasks directly below it, and renumbers the subsequent tasks. Project names the new task **<New Summary Task>**.

3. With **<New Summary Task>** selected, enter the **summary task name**.

To demote a summary task to a task

1. Select all subtasks below the summary task.
2. On the **Task tab**, in the **Schedule group**, click the **Outdent Task button**.

Enter milestone tasks

You might want to account for an important event for your project's plan, such as the end of a major phase of the project. To do this, you will create a milestone task.

Milestones are significant events that are either reached within the plan (such as the completion of a phase of work) or imposed upon the plan (such as a deadline by which to apply for funding). Because the milestone itself does not normally include any work, milestones are normally represented as tasks with zero duration. To visually distinguish milestones, their Gantt chart symbol appears as a diamond rather than a bar. However, you can flag any task of any duration as a milestone.

To enter a milestone task

1. In the **Task Name column**, click where you want to insert the milestone.
2. On the **Task tab**, in the **Insert group**, click **Milestone**.

Project inserts a new row for the new task and renumbers subsequent tasks. Project names the new task **<New Milestone>** and gives it a zero-day duration.

To mark a task of any duration as a milestone

1. Select a task name.
2. On the **Task tab**, in the **Properties group**, click **Information**.
3. In the **Task Information** dialog box, click **Advanced**.
4. Select the **Mark Task As Milestone** check box.

View the plan's critical path

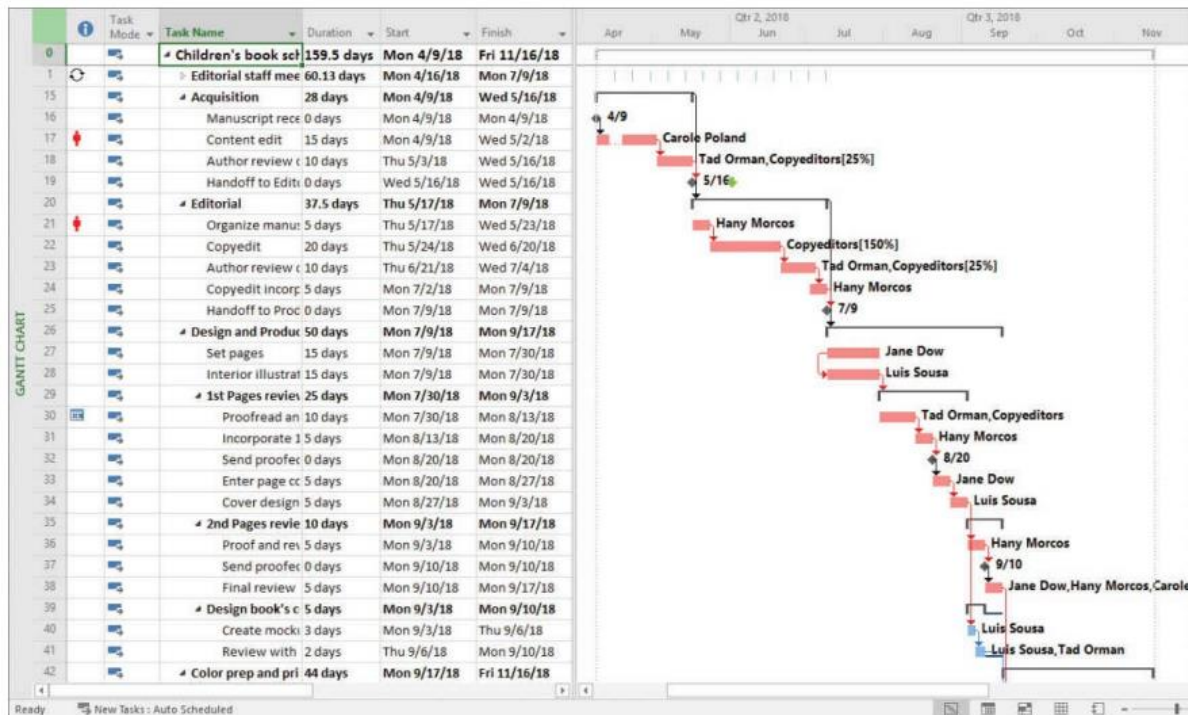
A critical path is the series of tasks that will push out the plan's end date if any of those tasks are delayed. The word critical in this context has nothing to do with how important these tasks are to the overall plan. It refers only to how their scheduling will affect the plan's finish date. However, the plan's finish date is of great importance to most projects.

If you want to shorten the duration of a plan to bring in the finish date, you must begin by shortening (also referred to as crashing) the critical path.

One key to understanding the critical path is to understand slack, also known as float. There are two types of slack: free and total. The amount of time a task can be delayed before it

delays another task is referred to as free slack. Total slack is the amount of time a task can be delayed before it delays the completion of the plan. (Note that after a task on the critical path is completed, it is no longer critical because it cannot affect the plan's finish date.)

A task is on the critical path if its total slack is less than or equal to a certain threshold—by default, if its slack is zero days (you can adjust this threshold if you want). In contrast, noncritical tasks have slack, meaning that they can start or finish earlier or later within their slack time without affecting the completion date of the plan.



As this Gantt chart view is formatted, both the tasks on the critical path (in red) and the free slack of noncritical tasks are displayed

Figure 10: Gantt chart with a critical path

To display the plan's critical path

1. On the **Format** tab, in the **Bar Styles** group, select the **Critical Tasks** check box. Project applies red formatting to the critical tasks' Gantt bars.

To display slack for noncritical tasks

1. On the **Format** tab, in the **Bar Styles** group, select the **Slack** check box. Project draws slack lines from the finish dates of the Gantt bars of the noncritical tasks.

To change the slack threshold for a task to be considered critical

By default, Project defines a task as critical if it has zero slack. However, you can change the amount of slack required for a task to be considered critical. You might do this, for example, if you want to more easily identify tasks that are within one or two days of affecting the project's finish date.

1. Click the **File** tab to display the **Backstage view**, and then click **Options**.
2. In the **Project Options** dialog box, click **Advanced**.

3. On the **Advanced** page, in the **Tasks Are Critical If Slack Is Less Than Or Equal To** box in the **Calculation options of this project** section, enter the number of days you want.

Setting up Resources

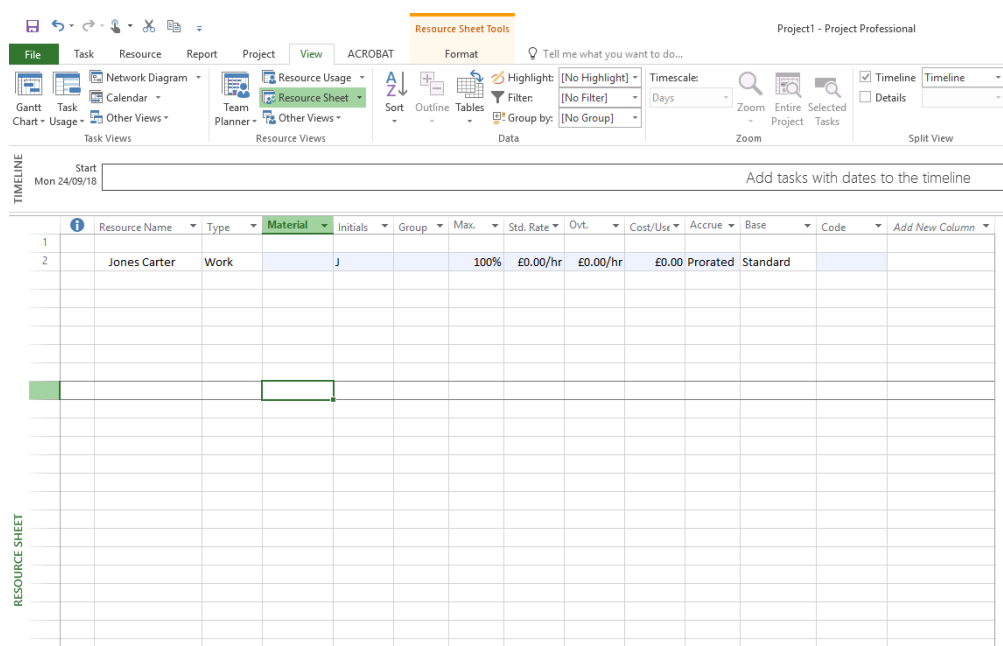
Resources include the people, equipment, and material needed to complete the work of a project. Effective resource management is one of the most significant advantages of using Project 2016.

You can manage three types of resources in Project, which are **work resources** and two special purpose resources: **cost** and **material**. **Work resources** include the people and equipment needed to complete the tasks in a project's plan. **Cost resources** represent categories of financial costs, derived from specific tasks that you need to account for in the plan.

Examples include categories of expenses like travel and entertainment. Material resources are consumables that get used up as the project proceeds. For example, a construction project might need to track steel or concrete as it is used throughout the project.

To enter work resource names

1. On the **View** tab, in the **Resource Views** group, click **Resource Sheet** to display the Resource Sheet view.
2. Click an **Empty cell** in the **Resource Name** column.
3. Enter your **Resource Names**, pressing the **Enter** key after each one. Project applies the default Work resource type in the Type field.



Resource Sheet

Figure 10: Resource Sheet with a single resource name

To insert a new work resource within a resource list

1. Click **Display** the Resource Sheet view.
2. Click in the cell in the **Resource Name** column where you want to insert the new resource.

3. On the **Resource tab**, in the **Insert group**, click **Add Resources**, and then click **Work Resource**.

Project inserts a new row with the **<New Resource>** name in the **Resource Name column**, and renumbers the subsequent resources.

4. With **<New Resource>** selected, enter the **Resource Name**, and then press **Enter**.

To delete a resource

1. **Right-click the resource name**, and then click **Delete Resource**

To adjust a resource's working capacity

1. On the **View tab**, in the **Resource Views group**, click **Resource Sheet**.

2. In the **Resource Sheet** view click the **Max. Units** field for the resource you want to adjust (the Max. portion of the field's column heading might be the only part visible).

To enter work resources pay rates

Almost all projects have some financial aspect, and cost limits the scope of many projects. Tracking and managing cost information in Project allows the project manager to answer such important questions as:

- What is the expected total cost of the project based on task durations and resource assignments?
- Is the organization using expensive resources to do work that less expensive resources could do?
- How much money will a specific type of resource or task cost over the life of the project?
- Is the organization spending money at a rate that it can sustain for the planned duration of the project?

In Project, you can enter standard rates and costs per use for work and material resources, in addition to overtime rates for work resources. Recall that in Project there are three types of resources: work, cost, and material.

You can specify when work resource standard and overtime costs should accrue. The options are at the start of work on a task, at the end of work on the task, or prorated (the default) evenly over the duration of the task.

In addition to or instead of cost rates, a resource can include a set fee that Project applies to each task to which the resource is assigned. This is called a cost per use. Unlike cost rates, the cost per use does not vary with the task's duration or amount of work the resource performs on the task.

	Resource Name	Type	Material	Initials	Group	Max.	Std. Rate	Ovt.	Cost/Use	Accrue	Base
1	Jun Cao	Work		J		100%	\$42.00/hr	\$63.00/hr	\$0.00	Prorated	Standard
2	Sharon Salavaria	Work		S		50%	100.00/wk	\$0.00/hr	\$0.00	Prorated	Standard
3	Toby Nixon	Work		T		100%	700.00/wk	\$0.00/hr	\$0.00	Prorated	Standard
4	Toni Poe	Work		T		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
5	Zac Woodall	Work		Z		100%	\$55.00/hr	\$0.00/hr	\$0.00	Prorated	Standard
6	Copyeditors	Work		C		400%	\$45.00/hr	\$0.00/hr	\$0.00	Prorated	Standard

View a resource's pay rates and other cost-related details in the Resource Sheet view

Figure 9: View of resources pay rate

To enter a standard rate for a resource

1. On the **View tab**, in the **Resource Views group**, click **Resource Sheet**.
2. In the **Resource Sheet view**, click the **Std. Rate field** for the resource for which you want to set a standard pay rate.

Depending on the width of the column, you might only see Std. in the column heading.

3. Enter a standard rate by **using the format of rate/period, such as 30/h.**

To enter an overtime rate for a resource

1. Click **Display** the **Resource Sheet view**.
2. Click the **Ovt. Rate field for the resource** for which you want to set an overtime pay rate.

Depending on the width of the column, you might only see Ovt. in the column heading.

3. Enter an overtime rate by **using the format of rate/period, such as 45/h.**

To enter a cost per use for a resource

1. Display the Resource Sheet view.
2. In the Cost/Use field, enter the value you want.

Setting up Cost Resources

Another type of resource you can use in Project is the cost resource. You can use a cost resource to represent a financial cost associated with a task in a plan. Although work resources (people and equipment) can have associated costs (hourly rates and fixed costs per assignment), the sole purpose of a cost resource is to associate a particular type of cost with one or more tasks. Common types of cost resources might include categories of expenses you want to track in a plan for accounting or financial reporting purposes, such as travel, entertainment, or training.

Cost resources do no work and have no effect on the scheduling of a task. The Max. Units, Standard and Overtime pay rate, and Cost Per Use fields do not apply to cost resources. After you assign a cost resource to a task and specify the cost amount per task, you can then see the cumulative costs for that type of cost resource, such as total travel costs in a project.

To add a cost resource

1. On the **View tab**, in the **Resource Views group**, click **Resource Sheet**.
2. In the **Resource Sheet view**, click in the **Resource Name column** where you want to add the cost resource.
3. On the **Resource tab**, in the Insert group, click **Add Resources**, and then click **Cost Resource**.

Project inserts a row for the new cost resource. Project names the new cost resource **<New Resource>**.

4. With **<New Resource >** selected, enter the cost resource name, and then press **Enter**.

To enter a note for a resource in the Resource Information dialog box

1. On the **View tab**, in the **Resource Views group**, click **Resource Sheet**.

2. Select a **Resource Name**.
3. On the **Resource Tab**, in the **Properties group**, click Notes.
4. In the **Notes box**, enter a note for the resource.

To enter a note for a resource by using the Resource Form

1. On the **View tab**, in the **Resource Views group**, click **Resource Sheet**.
2. On the **Resource tab**, in the **Properties group**, click the **Details button**.

In this type of split view, details about the item that is selected in the upper pane (a resource, in this case) appear in the lower pane. You can quickly change the selected resource name in the upper pane by clicking a name directly, or by clicking Previous or Next in the Resource Form. The Resource Form view can display one of several details; initially, it displays the Schedule details.

3. Click anywhere in the Resource Form view to change the focus to the Resource Form. The label of the Format tab changes to Resource Form Tools.

4. On the **Format tab**, in the **Details group**, click **Notes**.

The Notes details appear in the Resource Form view.

Type	Resource Name	Type	Material	Initials	Group	Max.	Std. Rate	Ovt.	Cost/Use	Accrue
1	Jun Cao	Work		J		100%	\$42.00/hr	\$63.00/hr	\$0.00	Prorated
2	Sharon Salavaria	Work		S		50%	100.00/wk	\$0.00/hr	\$0.00	Prorated
3	Toby Nixon	Work		T		100%	700.00/wk	\$0.00/hr	\$0.00	Prorated
4	Toni Poe	Work		T		100%	\$0.00/hr	\$0.00/hr	\$0.00	Prorated
5	Zac Woodall	Work		Z		100%	\$55.00/hr	\$0.00/hr	\$0.00	Prorated
6	Copyeditors	Work		C		400%	\$45.00/hr	\$0.00/hr	\$0.00	Prorated
7	Travel	Cost		T						Prorated

RESOURCE FORM

Name: Sharon Salavaria Initials: S Max units: 50% Previous Next

Costs

Std rate: \$1,100.00/wk Per use: \$0.00 Base cal: Standard

Ovt rate: \$0.00/h Accrue at: Prorated Group: Code:

Sharon's standard pay rate is adjusted for her half-time work schedule.

This is a great split view for quickly adding, editing, or reviewing resource details like notes

Figure 12: Note details and resource form

5. In the **Notes box**, enter the note for the resource.

Notice that as soon as you start typing the note, the Previous and Next buttons change to OK and Cancel.

6. Click **OK**.